

Polar Form

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Based on the lecture note uploaded on SPeCTRUM

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Learning Outcome

At the end of this lecture, you should be able to:

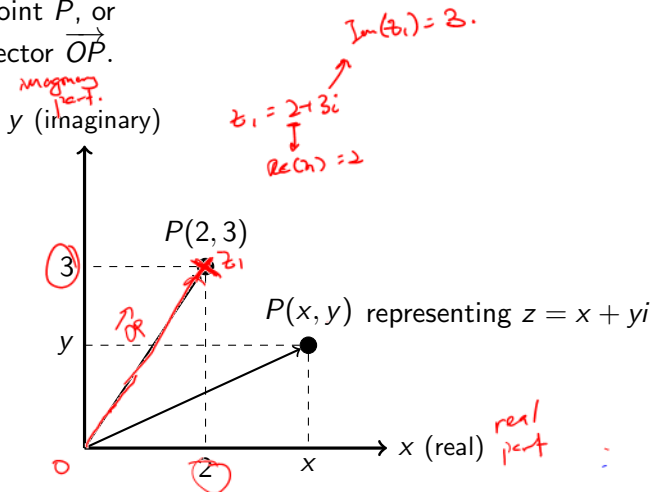
- Represent complex numbers in the Argand plane.
- Go back and forth between the polar form of a complex number and its cartesian form.
- Find modulus, argument and principal argument of complex numbers.
- Perform multiplication and division of complex numbers in polar form.
- Understand and apply the properties of modulus.

Argand Plane

Cartesian Form

Let $z = x + yi \in \mathbb{C}$ and P be the point (x, y) in the xy -plane. Then z can be represented by

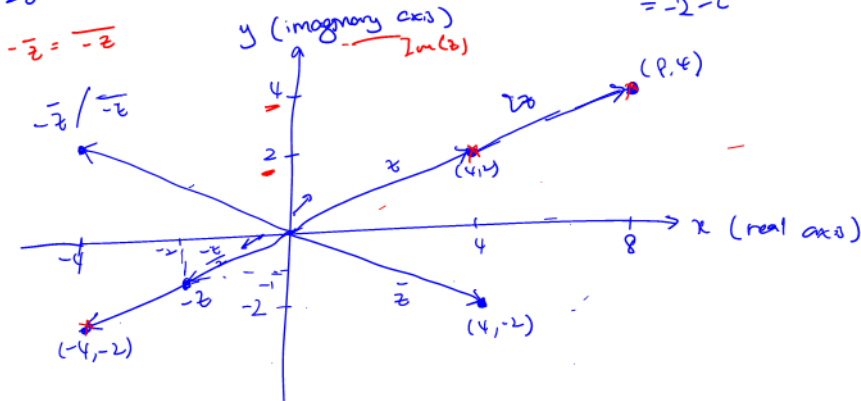
- 1 the point P , or
- 2 the vector $\overrightarrow{OP}$.



Argand Plane

Let $z = 4 + 2i$. Represent z , $\bar{z}$, $2z$, $-z$ and $\frac{-z}{2}$ in the Argand plane.

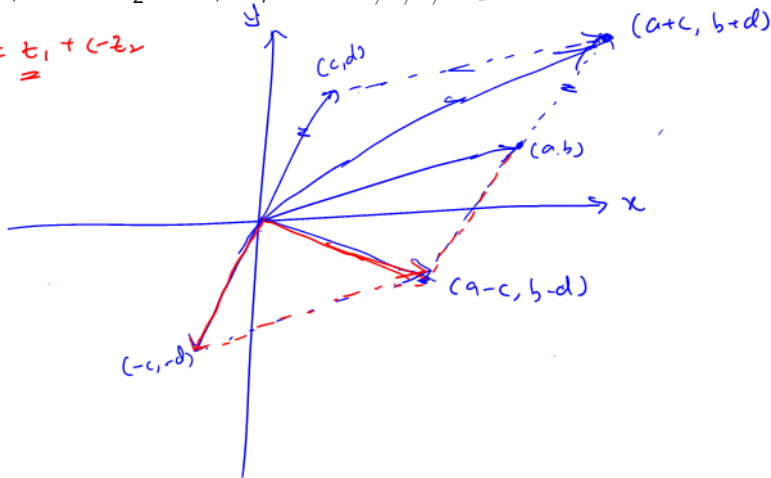
$$\begin{aligned}\bar{z} &= 4 - 2i & -z &= -(4 + 2i) & \frac{-z}{2} &= \frac{-4 - 2i}{2} \\ 2z &= 2(4 + 2i) = 8 + 4i & &= -4 - 2i & &= -2 - i\end{aligned}$$



Representation of Addition and Subtraction

Let $z_1 = a + bi$ and $z_2 = c + di$, where $a, b, c, d \in \mathbb{R}$.

$$z_1 - z_2 = z_1 + (-z_2)$$

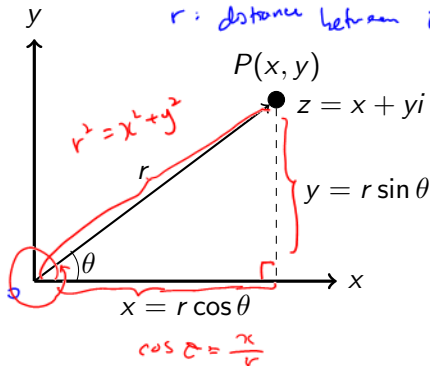


Polar Form

Cartesian form: $z = x + yi$, $x, y \in \mathbb{R}$

Polar form: $z = r(\cos \theta + i \sin \theta)$, where $r = \sqrt{x^2 + y^2}$, $\tan \theta = \frac{y}{x}$,
 $x = r \cos \theta$ and $y = r \sin \theta$.

θ : angle between positive x-axis and vector $\vec{OP}$ in anticlockwise.
 r : distance between O and P - / length of $\vec{OP}$



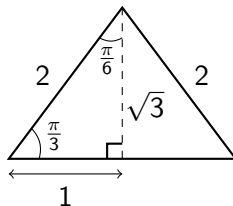
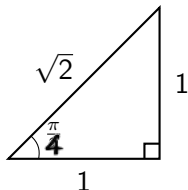
$$\sin \theta = \frac{y}{r}$$

$$\cos \theta = \frac{x}{r}$$

Some Special Angles

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	0	1
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined	0	undefined	0

θ in rad.



Modulus and Arguments

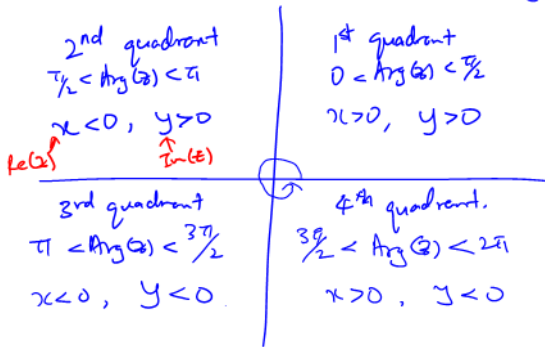
Modulus of z : $|z| = r = \sqrt{x^2 + y^2} \geq 0$

Argument of z : $\arg(z) = \theta = \arctan\left(\frac{y}{x}\right) = \theta + 2n\pi, n=0, 1, 2, \dots$
(Not unique)

Principal argument of z : $\text{Arg}(z) = \arg(z)$ in $[0, 2\pi)$ *← should only have one answer.*

Note: Some references might use the range $-\pi < \text{Arg}(z) \leq \pi$.

** In our course $0 \leq \text{Arg}(z) < 2\pi$*



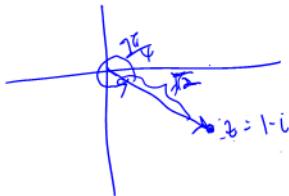
Modulus and Arguments

Example: Find the principal argument and the polar form of $z = 1 - i$

$$\tan \theta = \frac{-1}{1} = -1, \quad \theta = \frac{3\pi}{4} \text{ or } \frac{7\pi}{4}$$

X ✓

$x=1$
 $y=-1$
4th quadrant.



$$\theta = \arctan(-1) = \frac{7\pi}{4}$$

$$r = |z| = \sqrt{(1)^2 + (-1)^2} = \sqrt{2}$$

The polar form of $z = 1 - i$ is $z = \sqrt{2} \left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} \right)$

Modulus and Arguments

Example: Find the modulus and principal argument for the following

complex numbers: (i) $z_1 = \sqrt{3} + i$, 1st quadrant (ii) $z_2 = -\sqrt{3} + i$, 2nd quadrant
(iii) $z_3 = -\sqrt{3} - i$, 3rd (iv) $z_4 = \sqrt{3} - i$, 4th

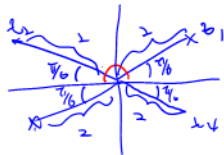
$$|z_1| = \sqrt{(\sqrt{3})^2 + (1)^2} = 2 = |z_2| = |z_3| = |z_4|$$

$$\text{Arg}(z_1) = \theta_1 = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

$$\text{Arg}(z_2) = \theta_2 = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

$$\text{Arg}(z_3) = \theta_3 = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$$

$$\text{Arg}(z_4) = \theta_4 = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$$



$$z_1 = 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right) \quad z_3 = 2\left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}\right)$$

$$z_2 = 2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right) \quad z_4 = 2\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$$

Multiplication in Polar Form

Let $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$

$$\begin{aligned} z_1 z_2 &= [r_1 (\cos \theta_1 + i \sin \theta_1)] [r_2 (\cos \theta_2 + i \sin \theta_2)] \\ &= r_1 r_2 [(\cos \theta_1 + i \sin \theta_1) (\cos \theta_2 + i \sin \theta_2)] \\ &= r_1 r_2 [\cos \theta_1 \cos \theta_2 + i(\cos \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2) \\ &\quad + \underline{i^2} (\sin \theta_1 \sin \theta_2)] \\ &= r_1 r_2 [\underbrace{(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2)} + i(\cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2)] \end{aligned}$$

$$\begin{aligned} * \cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B \\ \sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B \end{aligned}$$

$$z_1 z_2 = \underline{r_1 r_2} \left[\cos(\underline{\theta_1 + \theta_2}) + i \sin(\underline{\theta_1 + \theta_2}) \right]$$

$$|z_1 z_2| = r_1 r_2 = |z_1| |z_2|$$

$$\arg(z_1 z_2) = \theta_1 + \theta_2 = \arg(z_1) + \arg(z_2)$$

* Always check if $\arg(z_1) + \arg(z_2) > 2\pi$, we need to adjust it so that $0 \leq \arg(z_1 z_2) < 2\pi$

Division in Polar Form

Let $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$

$$\begin{aligned}\frac{z_1}{z_2} &= \frac{r_1(\cos \theta_1 + i \sin \theta_1)}{r_2(\cos \theta_2 + i \sin \theta_2)} \\&= \frac{r_1(\cos \theta_1 + i \sin \theta_1)}{r_2(\cos \theta_2 + i \sin \theta_2)} \frac{(\cos \theta_2 - i \sin \theta_2)}{(\cos \theta_2 - i \sin \theta_2)} \\&= \frac{r_1}{r_2} \left[\frac{(\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 - \cos \theta_1 \sin \theta_2)}{\cos^2 \theta_2 + \sin^2 \theta_2} \right]\end{aligned}$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$$

$$\left| \frac{z_1}{z_2} \right| = \frac{r_1}{r_2} = \frac{|z_1|}{|z_2|}$$

$$\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$$

Example

Let $z = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$ and $w = \sqrt{3}(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})$.

Find zw and $\frac{z}{w}$.

$$|z| = 2, \text{Arg}(z) = \frac{\pi}{3}$$

$$|w| = \sqrt{3}, \text{Arg}(w) = \frac{3\pi}{4}$$

$$\begin{aligned} zw &= (2)(\sqrt{3}) \left(\cos \left(\frac{\pi}{3} + \frac{3\pi}{4} \right) + i \sin \left(\frac{\pi}{3} + \frac{3\pi}{4} \right) \right) \\ &= 2\sqrt{3} \left(\cos \frac{13\pi}{12} + i \sin \frac{13\pi}{12} \right) \end{aligned}$$

$$0 \leq \frac{13\pi}{12} < 2\pi$$

$$\frac{z}{w} = \frac{2}{\sqrt{3}} \left(\cos \left(\frac{\pi}{3} - \frac{3\pi}{4} \right) + i \sin \left(\frac{\pi}{3} - \frac{3\pi}{4} \right) \right)$$

$$= \frac{2}{\sqrt{3}} \left(\cos \left(\frac{-5\pi}{12} \right) + i \sin \left(\frac{-5\pi}{12} \right) \right)$$

$$\frac{-5\pi}{12} \notin [0, 2\pi)$$

$$= \frac{2}{\sqrt{3}} \left(\cos \left(\frac{-5\pi}{12} + 2\pi \right) + i \sin \left(\frac{-5\pi}{12} + 2\pi \right) \right)$$

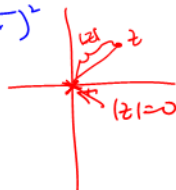
$$= \frac{2}{\sqrt{3}} \left(\cos \frac{19\pi}{12} + i \sin \frac{19\pi}{12} \right)$$

$$0 < \frac{19\pi}{12} < 2\pi$$

Properties of Modulus

Let $z_1, z_2, \dots, z_n$ be complex numbers. Then

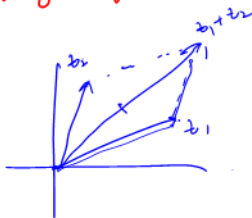
- $|z| \geq 0$ and $|z| = 0 \Leftrightarrow z = 0$.
 Let $z = x + yi$. $|z| = 0 \Leftrightarrow \sqrt{x^2 + y^2} = 0 \Leftrightarrow x^2 + y^2 = 0 \Leftrightarrow x = 0, y = 0 \Leftrightarrow z = 0$
- $|z| = |-z| = |\bar{z}|$.
 $(x + yi)(x - yi) = x^2 + y^2$
 $\Rightarrow \sqrt{x^2 + y^2} = |z|$
 $\Rightarrow |z|^2 = x^2 + y^2$
- $z\bar{z} = |z|^2 = |\bar{z}|^2$.
 $|z_1 z_2| = |z_1| |z_2|$
 $= (\sqrt{x^2 + y^2})^2 = x^2 + y^2 = |z|^2$
- $|z_1 z_2 \dots z_n| = |z_1| |z_2| \dots |z_n|$.
- $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}, z_2 \neq 0$
- $|z_1 + z_2 + \dots + z_n| \leq |z_1| + |z_2| + \dots + |z_n|$ (Triangle Inequality)
- $|z_1 \pm z_2| \geq ||z_1| - |z_2||$
- $z \neq 0 \rightarrow z^{-1} = \frac{\bar{z}}{|z|^2}$
 $\frac{1}{z} = \frac{1}{z} \cdot \frac{\bar{z}}{\bar{z}} = \frac{\bar{z}}{z \cdot \bar{z}} = \frac{\bar{z}}{|z|^2}$
 $= \frac{\bar{z}}{x^2 + y^2} = \frac{\bar{z}}{|z|^2}$
- $\overline{\bar{z}} = z$
 $z \cdot \bar{z}^{-1} = 1, z \neq 0$



$$|z_1 + z_2| \leq |z_1| + |z_2| \quad \leftarrow \text{triangle inequality.}$$

$$|z_1 + z_2 + z_3| \leq |z_1 + z_2| + |z_3|$$

$$\leq |z_1| + |z_2| + |z_3|$$



$$|z_1 - z_2| \geq ||z_1| - |z_2||$$

$$|z_1| = |(z_1 - z_2) + z_2| \leq |z_1 - z_2| + |z_2| \Rightarrow |z_1 - z_2| \geq |z_1| - |z_2|$$

$$|z_2| = |(z_2 - z_1) + z_1| \leq |z_2 - z_1| + |z_1| \Rightarrow |z_1 - z_2| \geq |z_2| - |z_1|$$

$$|z_1 - z_2| \geq -(|z_1| - |z_2|)$$

$$|z_1| - |z_2| \geq -|z_1 - z_2|$$

$$-|z_1 - z_2| \leq |z_1| - |z_2| \leq |z_1 - z_2|$$

$$|z_1 - |z_2|| \leq |z_1 - z_2|$$

$$-2 \leq x \leq 2$$

$$|x| \leq 2$$

$$|z_1 - z_2| \geq ||z_1| - |z_2||$$

$$\text{To show } |z_1 + z_2| \geq ||z_1| - |z_2||$$

$$|z_1| = |(z_1 + z_2) - z_2| = |(z_1 + z_2) + (-z_2)| \leq |z_1 + z_2| + |-z_2| = |z_1 + z_2| + |z_2|$$

$$|z_2| = |(z_2 + z_1) - z_1| = |(z_2 + z_1) + (-z_1)| \leq |z_2 + z_1| + |-z_1| = |z_1 + z_2| + |z_1|$$

Exercise : Complete this proof.